

NATIONAL CRITICAL INFRASTRUCTURE PROTECTION AGENCY

Incident Investigation Report: Operation BlackEcho (INC-88412)
TLP:AMBER | Public Sector Resilience Assessment

1. Executive Summary & Incident Timeline

On August 14, 2026, at approximately 02:41 UTC, the Central Operations Security Center (COSC) detected anomalous outbound traffic originating from core telemetry gateway servers in Cluster Delta-7. Subsequent forensic analysis confirmed unauthorized access by state-nexus threat group APT-39 (SilentHydra). The attack led to the compromise of 14 core supervisory control nodes and the exfiltration of 450 GB of operational telemetry archives.

Incident Identifier	INC-88412 (Operation BlackEcho)
Date of Incursion	August 14, 2026 (02:41 UTC)
Compromised Infrastructure	14 Telemetry & SCADA Gateway Servers
Data Exfiltrated	450 Gigabytes of telemetry archives
Service Interruption	4 hours 18 minutes partial telemetry blackout
Total Financial Exposure	\$2,850,000 USD (Remediation & Forensic Audits)

2. Technical Root Cause & Attack Vector Analysis

The initial entry point was traced to an external-facing edge bastion host running a vulnerable utility layer. Specifically, the adversaries exploited CVE-2024-3094 (liblzma/XZ Utils backdoor) to establish reverse SSH tunnels bypassing perimeter firewalls. Following initial persistence, the threat actors leveraged CVE-2024-21413 (Microsoft Outlook NTLM Credential Leakage) to harvest service account credentials belonging to svc_telemetry_admin. Privilege escalation was achieved within 38 minutes of landing on the internal subnet.

Indicator Type	Value / Artifact	Threat Action
IPv4 Address	198.51.100.47	C2 Command & Control Beacon
SHA-256 Hash	e3b0c44298fc1c149afb4c8996fb92427ae41e4...	Backdoored liblzma.so.5.6.0 binary
TLS Cert	CN=telemetry-gateway.internal.bad	Man-in-the-Middle proxy interception
File Path	/usr/local/bin/.blackecho_agent	Obfuscated persistence beacon script

3. Affected Systems & Quantitative Impact Assessment

The breach impact was concentrated across the eastern regional operational division. While safety-instrumented systems (SIS) remained isolated via hardware air-gaps, operational visibility was impaired across 3 regional dispatch centers.

Facility Name	Impacted Nodes	Outage Duration	Restoration Cost
Substation Apex-1	6 SCADA Nodes	4h 18m	\$1,120,000 USD
Telemetry Relay East	5 Distribution Relays	2h 45m	\$890,000 USD
Central Dispatch South	3 Gateway Servers	1h 10m	\$840,000 USD

4. Containment Actions & Strategic Hardening Mandates

The containment team executed emergency isolation protocols at 06:59 UTC on August 14, 2026. All egress routes to known command-and-control infrastructure were blocked at the national backbone BGP level.

Immediate Actions Completed:

1. Revoked all 84 kerberos service account tickets and enforced mandatory 24-character key rotation.
2. Decommissioned all 14 affected server instances and rebuilt systems from verified cryptographic golden images.
3. Patched CVE-2024-3094 across 100% of internal Linux nodes (verified 1,240 servers).
4. Enforced hardware FIDO2 MFA across all administrative ingress points.

Strategic Recommendations for Leadership:

- Mandate Zero Trust Micro-segmentation across OT/IT boundary layers within 60 days.
- Establish Immutable Blockchain Provenance logging for all operational telemetry artifacts.
- Allocate an emergency capital reserve of \$1,500,000 USD for continuous automated red-team simulations.